Math 242, S16 **Name:** Answer

Quiz 1 **SSN: xxx-xx-**

Instructions: **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (5 points) Given a first-order differential equation

$$y' = e^{-3y}.$$

Verify that $y(x) = \frac{1}{3}\ln(3x + C)$ defines a one-parameter family of solutions of the given differential equation, then determine a value of the constant C so that $y(x)$ satisfies the initial condition $y(1) = 0$.

$$y(x) = \frac{1}{3}\ln(3x+C) \Rightarrow y' = \frac{1}{3}\cdot\frac{1}{3x+C}\cdot 3 = \frac{1}{3x+C}$$

$$e^{-3y} = e^{-3\cdot\frac{1}{3}\ln(3x+C)} = e^{-\ln(3x+C)} = \frac{1}{3x+C}$$

$$\Rightarrow y' = e^{-3y}$$

$\Rightarrow y(x) = \frac{1}{3}\ln(3x+c)$ defines a one-parameter family of of solutions of $y' = e^{-3y}$

$$y(1) = 0 \Rightarrow \frac{1}{3}\ln(3\cdot 1 + c) = 0 \Rightarrow \ln(3+c) = 0$$

$$\Rightarrow 3 + c = 1$$

$$\Rightarrow c = -2$$

Problem 2. (5 points) Find a general solution for the first-order differential equation

$$\frac{dy}{dx} = \frac{1}{\sqrt[3]{2x+1}}.$$

$$y(x) = \int \frac{1}{\sqrt[3]{2x+1}}\,dx + C$$

$$= \int (2x+1)^{-\frac{1}{3}}\,dx + C$$

$$= \frac{3}{2}(2x+1)^{\frac{2}{3}}\cdot\left(\frac{1}{2}\right) + C$$

$$= \frac{3}{4}(2x+1)^{\frac{2}{3}} + C$$